

Data Engineering Project

1. Title: End-to-End E-Commerce Data Platform for Sales and Customer Analytics

2. Group Members:

- **Vishwa Narayanaswamy:**

Responsible for **data collection and data cleaning**. Prepares raw data, handles preprocessing, and ensures data quality before it enters the pipeline.

- **Sheerin Banu Shiek Mohamed:**

Responsible for **pipeline architecture and ETL workflows**. Builds and manages Airflow pipelines, Docker setup, deployment, and system monitoring.

- **Koushik Raj Singh:**

Responsible for **data analysis, warehouse design, and dashboard creation**. Designs SQL queries, builds warehouse tables, performs analysis, and develops dashboards for insights. Also contributes to documentation and testing.

3. Problem Statement:

E-commerce businesses generate large amounts of data, but without a proper system it is hard to use effectively. This project builds an automated platform to organize data and support smarter decisions.

4. Introduction:

E-commerce has become a major part of everyday life, with people shopping online for convenience and variety. Every search, click, and purchase creates valuable data that can help businesses understand customer behavior, track sales, and improve services.

However, in many organizations, data is stored in different systems such as order databases, customer records, and product catalogs. Because the data is scattered, it becomes difficult to combine and analyze it quickly. Many businesses still rely on manual reports using spreadsheets, which takes time and can lead to errors. As the amount of data increases, this approach becomes slow and inefficient.

This project focuses on building an End-to-End E-Commerce Data Platform to solve these problems. The system automatically collects data from multiple sources, cleans and transforms it, and stores it in an organized way. The processed data is then presented through dashboards and

reports that show insights like sales trends, customer behavior, and product performance. By automating the entire process, the platform ensures accurate and up-to-date information. Overall, this project shows how data engineering can turn raw data into useful insights that help businesses improve performance and make better decisions.

5. Project Execution

The project execution plan outlines the complete workflow and timeline for developing the End-to-End E-Commerce Data Platform for Sales and Customer Analytics. It helps organize project activities, manage task dependencies, track milestones, and ensure the project is completed efficiently and on schedule

- Gantt Chart

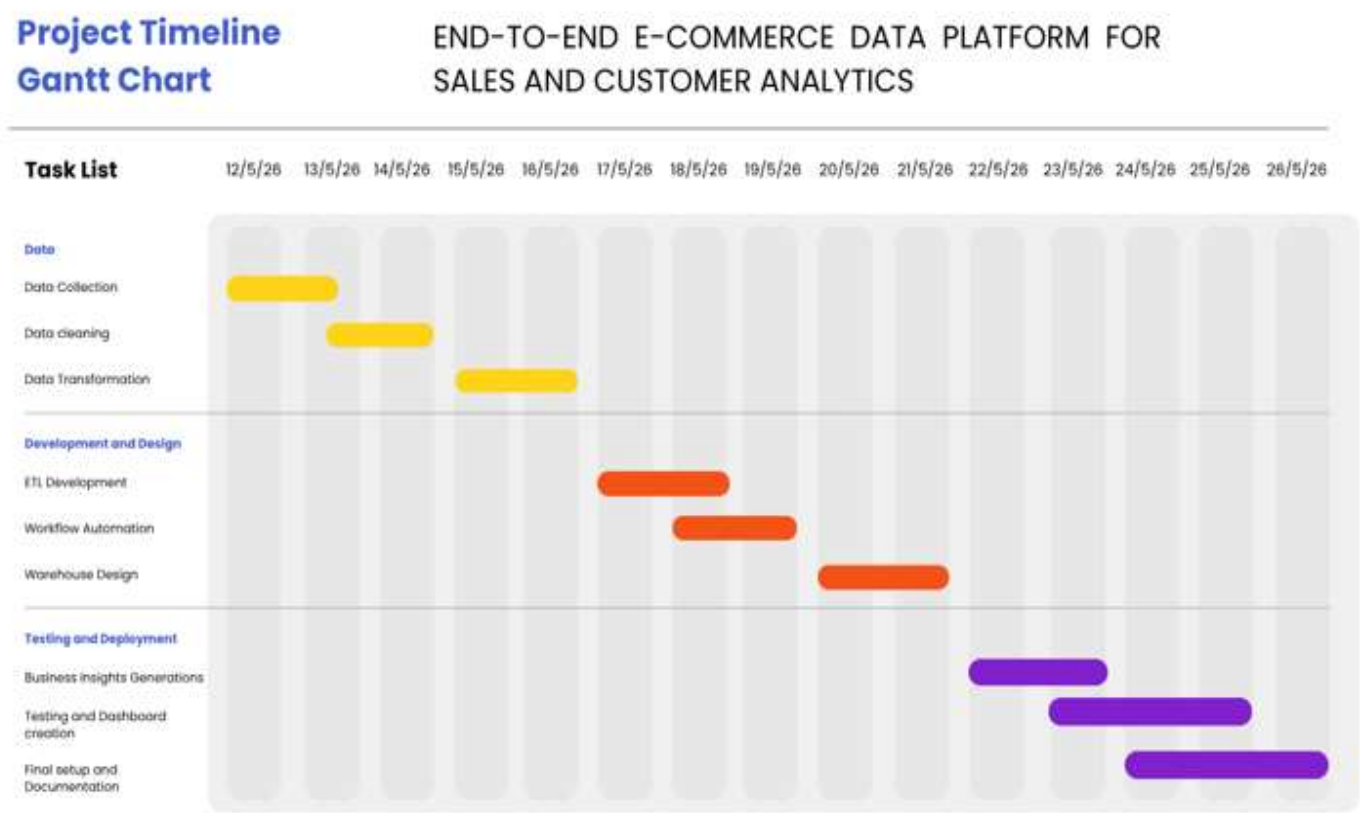


Fig 1: Gantt Chart

The Gantt chart gives a simple and clear view of how the End-to-End E-Commerce Data Platform for Sales and Customer Analytics project will be completed over 2 weeks. It shows when each major task, such as data collection, cleaning, ETL pipeline development, analytics, dashboard creation, testing, and deployment, will be carried out. This helps the team stay organized, track progress easily, and make sure the project is completed smoothly and on time.

- **Activity Flow Diagram**

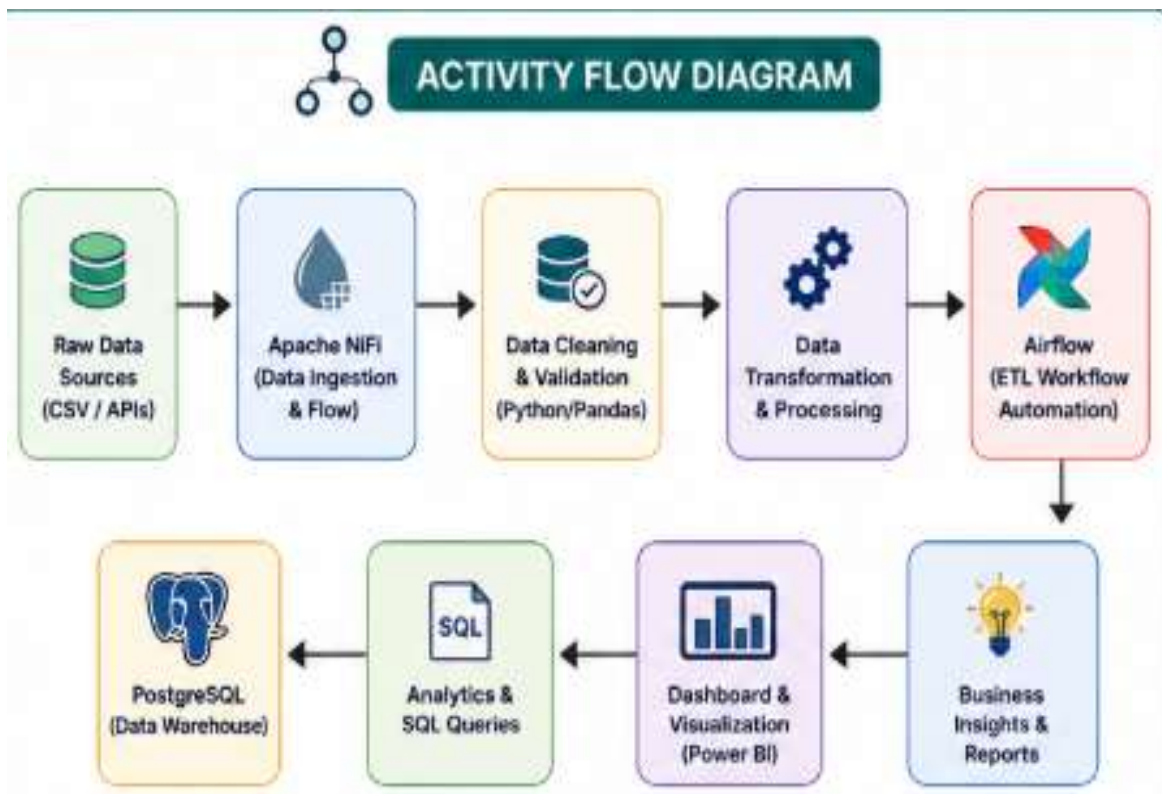


Fig 2: Activity Flow Diagram

The activity flow diagram illustrates the step-by-step workflow of the End-to-End E-Commerce Data Platform for Sales and Customer Analytics. It shows how raw e-commerce data is collected, processed, transformed, stored, and analysed through different stages to generate meaningful business insights and visual dashboards.

- **Milestones with Timelines**



The milestones with timelines section gives a clear breakdown of the important phases of the End-to-End E-Commerce Data Platform for Sales and Customer Analytics project and when each phase is expected to be completed. It helps in tracking progress step by step, from data collection and

processing to analytics, dashboard creation, testing, and final deployment, making the overall project execution more organized and easier to manage.

- **Directed Acyclic Graph (DAG)**

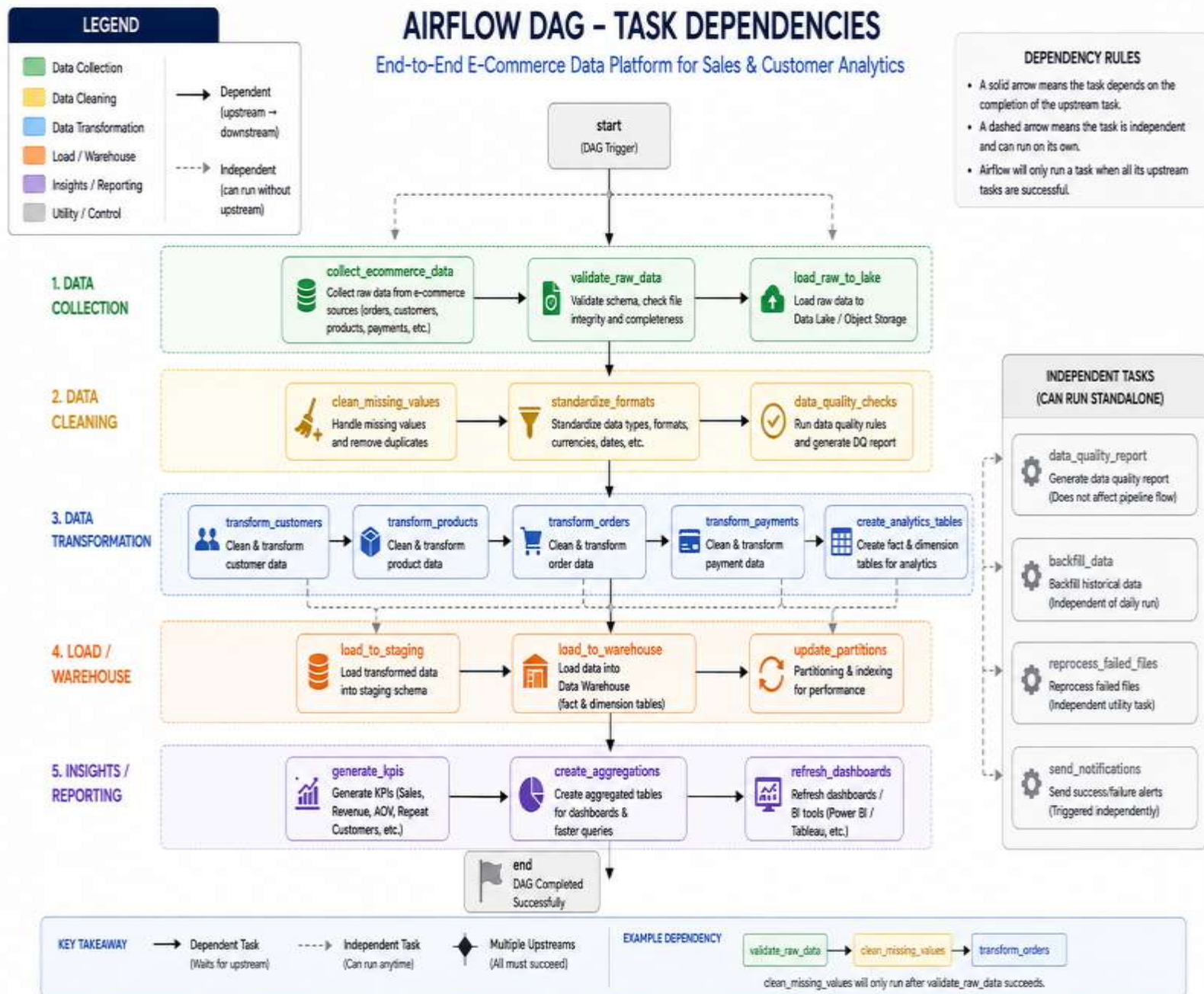


Fig 3: DAG

In the Directed Acyclic Graph (DAG) for the end-to-end e-commerce data platform, the core pipeline consists of dependent tasks that must execute in a fixed sequence because each task relies on the output of the previous one. The dependency flow begins with Data Collection, followed by Data Cleaning, Data Transformation, ETL Development, Workflow Automation, Warehouse Design, Business Insights Generation, Testing and Dashboard Creation, and finally Final Setup and Documentation. For example, Data Cleaning cannot start until the raw data has been collected, and Business Insights Generation cannot begin until the data warehouse has been populated. In contrast, independent tasks such as generating the Data Quality Report, sending email notifications, monitoring pipeline execution, and reprocessing failed records do not block the main workflow and can run separately or in parallel after specific stages complete. This distinction helps clarify which tasks are critical path activities and which are supporting tasks within the DAG.

6. Tools and Technologies

- **Python**

Used for data cleaning, preprocessing, ETL scripting, and automation tasks.

- **Pandas & NumPy**

Used for handling datasets, performing data transformation, and analysis.

- **Apache NiFi**

Used for data ingestion, data routing, and managing the movement of raw data between different systems.

- **Apache Airflow**

Used for workflow scheduling, DAG management, and automating ETL pipelines.

- **Apache Spark**

Used for scalable and large-scale data processing and analytics.

- **PostgreSQL**

Used as the data warehouse for storing cleaned and transformed data.

- **SQL**

Used for querying the database and generating business insights.

- **Docker**

Used for containerization and maintaining a consistent development environment.

- **Power BI / Tableau**

Used for creating dashboards and visualizing sales and customer analytics.

- **Git & GitHub**

Used for version control, collaboration, and project repository management.

7. Cost Management

The End-to-End E-Commerce Data Platform for Sales and Customer Analytics project is designed in a cost-effective manner by mainly using free and open-source tools such as Python, Apache NiFi, Apache Airflow, PostgreSQL, Docker, and GitHub. Since these technologies are freely available, the project can be developed and tested on a normal personal computer without significant software or infrastructure costs. The overall setup mainly requires a system with moderate RAM, storage, and internet connectivity. By using workflow automation and open-source technologies, the project achieves efficient data processing and management while keeping the overall cost practical and affordable.